MILITARY SPECIFICATION

LOADER, SCOOP TYPE, DED, 4 X 4, ARTICULATED FRAME STEER, 2-1/2 CU. YD., COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a commercial model, articulated frame steer, four wheel drive, diesel engine driven, 8,000 pound operating load capacity loader with a 2-1/2 cubic yard, multipurpose, front-end mounted, scoop-type bucket.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-3351	- Tractor, Full-Tracked, Low Speed; Tractor, Wheeled, Agricultural; and Tractor, Wheeled, Industrial; and their Attachments, Packaging of.
MIL-M-7866	- Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-I-14511	- Insulation, Sheet, Cellular, Plastic: Thermal.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Artic.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits For Army Materiel.
MS35424	- Light, Marker, Clearance-Blackout, 24-Volt.
MS51318	- Headlight, Blackout, 24-Volt, Waterproof.
MS51330	- Stop Light - Tail-Light, Vehicular, 24-Volt, Blackout Stop.

HANDBOOK

MILITARY

AIR FORCE SYSTEM COMMAND Design Handbook DH 1-11	- Air Transportability.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawing, and publications form a part of this specification to the extent specified herein.

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to the US Army Mobility Equipment Research and Development Command, ATTN: DRDME-H, Fort Belvoir, VA 22060.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive. Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The loader shall be the latest model of the standard product of the contractor. The loader shall have been marketed and in commercial field use in significant numbers equipped with a multipurpose bucket in the 2-1/2 cubic yard size for at least 1 year prior to the invitation for bids or request for proposal. Product improvements incorporated in the commercial production units during this 1 year period are acceptable. When specified requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment shall be furnished. The loader shall be equipped with any optional equipment necessary to comply with requirements specified herein. Optional equipment is defined as equipment not standard with the loader but which has been furnished to the commercial customer, such as special features or allied equipment. The loader shall be equipped with all components necessary to enable the loader to function reliably and efficiently in sustained operation. The loader shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect at the time of invitation for bids or request for proposal. For definition of terms used herein, SAE J732 shall apply.

3.2 Description. The loader shall be rubber-tired, diesel-engine driven four wheel drive with rear axle oscillation and shall have an articulated frame steering system. The loader shall be equipped with a hydraulically operated, front mounted, scoop type, multipurpose bucket (see 3.19.1).

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3.3 First article (preproduction model). The contractor shall furnish one loader for examination and demonstration within the time frame specified (see 6.2), to prove, prior to starting production, that his production methods and choice of design detail will produce loaders that comply with the requirements of this specification. Examination and demonstrations shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). At the option of the contractor, two loaders may be furnished to expedite examination and demonstration.

3.4 Production loader. All loaders furnished under this specification shall be physically and mechanically identical to the first article loader that is approved by the Government, except that product improvements incorporated in the contractor's standard product lines are permitted at no cost to the Government subject to the following:

(a) Physical and performance characteristics have not been degraded.
(b) New components involving Government selected repair parts are directly interchangeable with the superseded component or part.
(c) The Government is notified of all changes to any item on the recommended parts list where fit and function is effected a minimum of 45 days prior to the production date of the change. When the changes and alterations have been incorporated in the production loader, the Government shall be furnished the production effectivity by serial number and dates.
(d) Appropriate changes are made to the Government validated operating, maintenance and parts manuals before delivery.
(e) The contractor shall be responsible to provide repair parts for the loader for the service life of the loader.

3.5 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.6 Safety. The loader shall conform to all Federal Laws and regulations governing air-pollution and other health and safety requirements which are in effect for this type of equipment at the time of manufacture. All available standard or optional accessories shall be included to provide protection for maintenance and operating personnel from rotating or reciprocating parts from all parts subject to high operational temperatures and so located as to be a hazard. Exhaust discharge shall be directed so as to not endanger personnel. All platforms and steps shall have anti-skid surfaces. Contractor's standard grab irons shall be provided. A decal shall be furnished and shall be located in the area of the articulation point, warning of limited personnel clearance during frame articulation and placement of steering frame lock for the loader operating and transport modes.

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3.6.1 Sound level. The loader shall be equipped with all available standard and optional sound suppression equipment. If the sound level in the area occupied by the operator exceeds category D of MIL-STD-1474 when measured in accordance with SAE J919, the contractor shall perform an evaluation of the feasibility of lowering the sound level to 85 dB(A) or less and provide an analysis and cost estimate of meeting the 85 dB(A) limit. If the Government concludes that it is unjustifiable and unfeasible to require the 85 dB(A) limitation, the contractor shall provide a caution plate and appropriate warning in the maintenance manual in accordance with MIL-STD-1474. If the sound level with cab doors open exceeds 85 dB(A), when it is 85 dB(A) or less with the doors closed, a caution plate shall be provided stating: "CAUTION, HEARING PROTECTION REQUIRED WHILE OPERATING WITH DOORS OPEN". The caution plate shall have a yellow background with black lettering and shall be made of a corrosion resistant material. The caution plate shall be permanently affixed to the loader in a conspicuous, protected location in the operator's compartment. The exterior sound level of the loader shall not exceed 88 dB(A) when measured in accordance with SAE J88.

3.6.2 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The loader shall be equipped with a bolted down or pin connected ROPS/FOPS structure which may be integral with the fully enclosed cab (see 3.15). The structure shall be removable and replaceable without cutting or welding. The ROPS/FOPS shall conform to SAE J1040 and SAE J231, respectively. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. Quick disconnect fittings (electric, etc) shall be provided as applicable to facilitate ROPS/FOPS structure (ROPS cab) removal. A slinging eye(s) conforming to MIL-STD-209 shall be provided as attaching points for ROPS/FOPS structure (ROPS cab) removal.

3.6.3 Fenders. Front and rear fenders shall be furnished. Rear fenders may be an integral part of the rear housing or structure.

3.6.4 Horn. A conventional operator controlled horn shall be furnished.

3.6.5 Backup alarm. A backup alarm shall be provided on the loader and shall comply with SAE J994 type C or D. The backup alarm signal shall be audible above, and distinguishable from the surrounding noise level.

3.7 Physical characteristics.

3.7.1 Weight. The fully serviced and fueled loader equipped as specified herein with bucket, ROPS/FOPS and cab, but with any add-on type rear counterweight removed, shall weigh not less than 25,000 pounds. The fully serviced and fueled loader equipped with bucket, ROPS/FOPS and cab and with all rear counterweight installed, shall weigh not more than 32,000 pounds.

3.7.2 Dimensions. The loader shall conform to the dimensions shown in table I. (Reference SAE J732 - a through g).

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TABLE I. Dimensions.

Characteristic	Requirement
(a) Height to hinge pin at full lift, minimum.	11'8"
(b) Overall height, maximum.	11'6"
(c) Overall length, maximum.	26'
(d) Forward dumping dump height measure to the bucket cutting edge at full lift and 45 degree dump angle minimum.	8'4"
(e) Forward dumping reach measured to bucket cutting edge at full lift and 45 degree dump angle, minimum.	2'7"
(f) Maximum rollback at carry postion, minimum.	40 degrees
(g) Digging depth, minimum.	1.5"
(h) Ground clearance, minimum.	1'2"
(i) Maximum clamshell opening, minimum.	3'10"
(j) Bucket as a dozer, depth of cut with pitch angle of dozer cutting edge 50 degrees from the horizontal minimum.	2"
(k) Rear axle oscillation, (unless articulation joint allows for oscillation), each direction, minimum.	10-1/2 degrees

3.8 Use. The loader and attachments shall withstand the usage encountered in military operations such as loading trucks from stockpiles, stockpiling material, and excavating undisturbed and compacted soil without breakage, malfunction or permanent deformation. The loader shall also be used as a clamshell to handle irregularly shaped objects as a dozer for general bulldozer work and as a scraper.

3.9 Environmental conditions. Without use of the any external means, equipped with only standard or optional equipment, the engine shall start within five minutes and the loader shall operate under full-load conditions within fifteen minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be integral with the loader and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The loader shall be capable of operating in hot, humid tropical environments and dry, dusty environments. The loader shall be capable of being stored for extended periods in temperatures ranging from mimus 30 to plus 160 degrees F without damage.

3.10 Performance. The loader shall have the following performance capabilities:

(a) Lift, lower, and dump rated loads.
(b) Load the bucket to the SAE rated heaped capacity (reference SAE J742).
(c) Load stockpiled material into dump trucks.
(d) Dig and transfer stockpiled materials.
(e) Excavate undisturbed and compacted soil.
(f) As a clamshell, the loader shall handle irregularly shaped objects.

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(g) As a dozer, the loader shall perform backfilling and general bulldozer work. The dozer segment shall not interfere with the digging position of the bucket when bank loading with the bucket cutting edge horizontal.

(h) As a scraper, the loader shall load the bucket to its SAE struck capacity in undisturbed clay-loam soil (reference SAE J742).

3.10.1 Operating load. The loader shall have an SAE rated operating load of 8,000 pounds (reference SAE J818).

3.10.2 Tipping load. The loader shall have a full turn tipping load, SAE rating, of not less than 16,000 pounds (reference SAE J732).

3.10.3 Lift capacity. The loader shall have an SAE lift capacity at the carry position of not less than 20,000 pounds. The loader shall have an SAE lift capacity to maximum height of not less than 10,000 pounds (reference SAE J732).

3.10.4 Breakout force. The loader shall have an SAE breakout force of not less than 17,500 pounds (reference SAE J732).

3.10.5 Clam closure force. The closure force at the lip of the closed clam shall be not less than 7,000 pounds.

3.10.6 Drawbar pull. The empty loader shall develop a drawbar pull of not less than 16,000 pounds at 1.5 mph, at the specified operating weight, on a dry level, concrete surface, with not more than a 15 percent slip factor.

3.10.7 Travel speed. The empty loader shall be capable of a forward travel speed of not less than 19 mph on level payment.

3.10.8 Cycle time. The loader shall have an SAE raising time of not more than 7.5 seconds, and SAE lowering time of not more than 5.5 seconds, and an SAE dump time of not more than 4.0 seconds (reference SAE J732).

3.10.9 Clearance circle. The loader shall have an SAE clearance circle diameter of not more than 45 feet, 1 inch (reference SAE J732).

3.10.10 Stability. The loader shall negotiate 27 percent side slopes and 50 percent longitudial slopes without any wheel of the loader losing contact with the ground. All parts requiring lubrication shall be lubricated when the loader is operating under these conditions.

3.10.11 Fording. The loader shall start and operate during and after fording operations in fresh water to any depth up to 20 inches including wave action without damage and without leakage of water into reservoirs containing lubricants.

3.11 Engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The diesel engine shall be the engine furnished commercially including all systems, components, accessories, and

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auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow. The engine shall be the same engine that has been used in the manufacturer's standard commercial loader of the same size for at least one year prior to contract award and shall be of USA manufacture.

3.11.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816, shall be not less than 130 horsepower.

3.11.2 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.9 shall be provided.

3.11.3 Fuel tank. The fuel tank shall be of a type normally furnished as standard equipment with a capacity for 10 hours of continous normal operation. The tank shall be equipped with a means for draining water and sediment from the bottom of the tank.

3.11.4 Engine side panels. The loader shall be equipped with removable engine side panels with quick-disconnect latches or horizontally or vertically hinged engine compartment doors or a hinged "tilt away" type engine enclosure.

3.11.5 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission and shall have the capability of closing automatically when released. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.12 Drive train.

3.12.1 Transmission. A full power shift transmission shall be furnished. The transmission shall provide not less than three forward speeds and three reverse speed or when a dual range, twin turbine torque converter is furnished not less than two forward gears and one reverse gear resulting in four forward speeds and two reverse speeds. The loader shall be equipped with a transmission disconnect that automatically disconnects the drive train when the service brakes are applied. The loader shall also have the capability of having the service brake applied without disconnecting the drive train. The starting system shall be such that the transmission must be placed in neutral before the engine can be started or the transmission must be placed in neutral before the loader can be set in motion.

3.12.2 Torque converter. The loader shall be equipped with a torque converter. A torque converter oil cooler shall be furnished.

3.12.3 Drive axles. The front and rear drive shall have differentials and planetary final drives. A limited slip differential or a hydraulic differential

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lock shall be furnished in the front or rear axle or torque transferring differentials of the torque bias type, in the front and rear axles or front axle only.

3.13 Tires. Tires shall be of the traction type tread (L-2), nylon, tubeless, with not less than a 12 ply rating. Tires shall be of the wide base type with a size of 20.5 x 25. All tires shall be of the same size, type, and ply rating.

3.14 Steering and braking.

3.14.1 Steering system. The loader shall have an articulated frame type steering system with a minimum of 30 degrees of articulation in each direction. The steering system shall provide full control of the loader under all normal operating conditons and shall provide emergency steering conforming to SAE J53 without engine power.

3.14.2 Braking system. A foot controlled service braking system shall be provided with disc brakes acting on all four wheels. A parking system and an emergency stopping system shall be provided. The braking system shall comply with SAE J1152. When nonpetroleum based hydraulic brake fluid is used in a hydraulic braking system, it shall be of the silicone type in accordance with MIL-B-46176.

3.15 Cab. A fully enclosed removable operator's cab (or ROPS cab) with door(s) windows, and front and rear windshield shall be furnished. The door(s) of the cab shall have a latch mechanism that will latch the door(s) in the full open position. Front windshield wiper(s) and windshield washer system shall be furnished. The cab shall be insulated against cold and shall be sound suppressed. A heater shall be furnished and shall have a rating of not less than 18,000 Btu/hr. Ventilation fans or defrosters shall be furnished that direct airflow toward all glazing. Each fan shall have an "on-off" switch and be directionally adjustable. Quick disconnect fittings (electric, etc) shall be provided as applicable to facilitate cab removal. The cab shall have slinging eye(s) to aid in removing the cab. Outside mirrors mounted on left and right side of the cab shall be furnished.

3.15.1 Seat. The contractors standard commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone and all instruments and gages mounted on the instrument panel, when seated with seat belt fastened and properly adjusted.

3.15.2 Instruments and indicators. The following instruments and indicators shall be provided:

(a) Engine oil pressure gage, warning light, or diode monitoring system.
(b) Engine coolant temperature gage, warning light, or diode monitoring system.
(c) Torque converter oil temperature gage, warning eight, or diode monitoring system.

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(d) Ammeter or voltmeter, warning light, or diode monitoring system.
(e) Engine hourmeter or service meter.
(f) Air pressure gage (when equipped with air system).
(g) Low air pressure warning buzzer (when equipped with air system).
(h) Fuel level indicator (dipstick or gage).
(i) Hydraulic oil level indicator (dipstick, sight tube, or gage).
(j) Air cleaner restriction indicator.

Additional instruments and indicators standard with the contractor's commercial practice, which contribute to the safety and reliability of loader operation, shall be furnished.

3.15.3 Controls. All controls for the loader shall be located in the operator's compartment and shall be within the reach of the operator in his normal operating position. All controls shall be identified by decals or plates on/or adjacent to each control to indicate the function or use.

3.16 Hydraulic system. The loader shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all attachments and related components. An adjustable automatic boom kickout and an automatic bucket positioner shall be furnished.

3.17 Electrical system. The loader shall be equipped with the contractor's standard or optional electrical system which meets the requirements specified herein. A 24-volt starting system shall be furnished if available as standard or as an option.

3.17.1 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.9.

3.17.2 Alternator. The contractor's high output option alternator shall be furnished. The rated output shall be not less than 50 amps.

3.17.3 Lighting. The following minimum lights with associated circuits, circuit breakers, or other protective devices and switches shall be provided:

(a) Four front work/travel lights.
(b) Two combination stop and taillights.
(c) Two rear work lights.
(d) Turn signals with emergency flashers (if offered).
(e) Instrument panel lighting.

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3.17.4 Blackout lighting. A blackout lighting system shall be furnished to meet the following requirements:

(a) Each vehicle will be equipped with the following blackout lights:
  (1) One left front corner marker light conforming to MS51318.
  (2) Two front corner marker lights conforming to MS35424-1.
  (3) Two rear corner marker and stop/taillights conforming to MS51330.

(b) The blackout system will be controlled by an interior switch, readily accessible to the driver, which will prevent accidental disengagement of the blackout system from the blackout mode.

(c) All regular exterior lights and horn, if electric, will be made inoperable automatically when in the blackout mode.

(d) The interior lighting will have a variable brightness control switch readily accessible to the driver.

(e) A separate wiring harness shall be provided for the blackout lights only if the required blackout lights cannot be powered using the standard wiring harness.

3.17.5 Radio interference suppression. The loader shall be suppressed to limit electromagnetic radiation to limits of SAE J551. Suppressors, when used, shall conform to SAE J552.

3.18 Towing device. The loader shall be equipped with a fixed type drawbar including a drawbar pin and a positive pin retaining means.

3.19 Attachments.

3.19.1 Multipurpose bucket. The loader shall be furnished with a multipurpose type bucket. The SAE nominal heaped bucket capacity rating shall be 2-1/2 cubic yards when rated in accordance with SAE J742. The bucket shall be equipped with not less than eight bolt-on teeth. The bucket width shall be at least 1 inch greater than the loader without bucket. The bucket shall operate as a front loader, clamshell, dozer, and scraper.

3.19.2 Counterweight. Removable rear counterweight may be furnished on the loader to improve stability and to meet the requirements of this specification. The counterweight furnished shall be the same as that furnished by the contractor on his standard commercial loader when equipped with a 2-1/2 cubic yard multipurpose bucket. Tire ballast is not acceptable and shall not be furnished.

3.20 Transportability. The loader shall be capable of being transported by truck trailers, rail, ocean-going carriers, and C130, C141 and C5A aircraft and withstand the impact forces encountered in shipment without damage or permanent deformation. The loader shall be equipped with slinging provisions and shall be capable of being lifted without damage using slings. Tiedown provisions shall be furnished for transport by the above modes. A steering frame lock in accordance with SAE J276 shall be provided. Complete diagrams and instructions for lifting the loader and tying down the loader on freight cars and truck

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trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport.

3.20.1 Air transportability. The contractor shall provide sufficient data to the Government to confirm that the loader being supplied can be loaded and transported in the C-130, C-141 and C-5A aircraft. Cargo envelopes and allowable axle weights for transport in these aircraft, except axle weight for C-141, are found in AFSC Design Handbook DH1-11. The following axle weights and minimum distances between axles for transport in the C-141 are: (a) Two 10,000 pound axles spaced 4 feet, (b) Two 15,000 pound axles spaced 10 feet, (c) One 10,000 pound axle and one 15,000 pound axle spaced at least 8 feet. Disassembly to achieve air transportable configuration shall be accomplished in not more than two hours using only common tools and available lifting devices. The loader shall be capable of being driven on and off the aircraft in the transportable configuration. All removed parts shall be pallitized for air transport with the loader. The loader shall be provided with a minimum of 1/4 of its fuel capacity in the tank.

3.21 Slinging provisions. The loader shall be provided with four slinging provisions conforming to MIL-STD-209. The provisions shall enable the loader to be lifted in the normal traveling or operating position. Provisions shall be positioned so that the lifting strain shall be in line with the longitudinal axis of the eye of the provisions. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure or permanent deformation of the loader or the provisions.

3.22 Tiedown provisions. The loader shall be provided with integral provisons to permit tiedown of the loader to the floor or deck of the transportation medium. The tiedown provisions shall conform to MIL-STD-209 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the loader or the provison.

3.23 Shipping data plates. Shipping data plates for each loader shall be furnished and shall show the silhouette of the loader indicating the location and capacity of slinging provisions, tiedown provisions, the center of balance and shipping weight. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. The plates shall be attached to the left side of the loader and shall show the location as viewed from the left side.

3.23.1 Vehicle Classification data. The contractor shall complete and submit the vehicle classification data sheet, figure 1, within the time frame specified (see 6.2).

3.24 Stenciling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown and slinging provisions, tire pressure and the center of gravity.

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3.25 Identification plates. Each loader shall be furnished with an identification plate showing the contractor's model number, NSN, serial number, USA registration number, date of manufacture, and date of delivery. The identification plate shall be permanently affixed to the loader in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer.

3.26 Instruction plates. The loader shall be equipped with instruction plates including warnings and caution suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the loader. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer.

3.27 Treatment and painting. The portions of the loader normally painted shall be cleaned, treated, primed and painted in accordance with the contractor's standard commercial practice. The finishing coat shall be in accordance with MIL-E-52798, MIL-E-52835, or MIL-E-52929, forest green color.

3.28 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each loader shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924 and MIL-G-23827 as applicable.

3.29 Technical publications. Such technical publications as are specified (see 6.2), shall be furnished.

3.30 Special tools. Any nonstandard or special tools required for equipment operation and operator maintenance of the loader shall be furnished with the loader.

3.31 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.32 Technical information package.. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2), depicting the loader and its components exactly as it will be furnished. The Technical Information Package shall be submitted with all bids and proposals.

3.33 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.3).

(a) Container NSN 7520-00-559-9718 complete with maintenance forms. (The contractor shall furnish the fasteners and install the containers on the loaders).
(b) Vehicle weight classification sign set, NSN 9905-00-565-6267.

3.34 Basic issue item. When specified (see 6.2), each item in the Basic Issue Item List shall be furnished with each loader.

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3.35 Winterized loader. When specified (see 6.2), a quantity of loaders as specified (see 6.2) shall be winterized to permit starting and operation in arctic climates at ambient temperatures to minus 50° F and storage at ambient temperatures to minus 65° F. The contractor shall furnish one loader for examination and demonstration with the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce winterized loaders that comply with the requirements of this specification.

3.35.1 Winterization kit. The kit shall consist of the components for preheating loader engine coolant and battery box(s); also insulated sound suppressed ROPS cab, hot water personnel heater (equipped with one or more shut-off valve(a) in the coolant lines to stop and to vary the flow of coolant to the heater, defrosting system for cab windows, front and rear windshield wipers, radiator winterfront, insulated engine and radiator compartments, insulated battery box(s), battery finned connectors, and an alcohol evaporator when air brakes are furnished. Winterization components shall not affect the performance of the loader and all other loader requirements shall be met as specified herein.

3.35.2 Heater and heat exchanger. The Eberspacher "Swingfire" heater, Part No. 26-9002, shall be utilized for preheat purposes. When specified (see 6.2), a quantity of Swingfire heaters as specified (see 6.2) shall be furnished. The Eberspacher heat exchanger (water jacket), Part No. 26-9013, compatible with the Swingfire heater shall be provided for heating the engine/coolant. The heat exchanger shall be located near the engine compartment for easy accessibility for insertion of heater. The heater unit starting operation shall be provided ty the loader's 24-volt electrical system using a wiring harness with Bosh Receptacle, Part No. 135050012, compatible with the heater starter cable connector. After heater start and insertion into heat exchanger, preheating of the engine coolant shall be provided by thermo-siphon action. The heater exhaust hot air shall be routed to the battery box(s) for hot air drenching evenly of the battery box(s) and then routed away from battery box(s) to the atmosphere. The hot air flow shall not be impeded by excessive back pressure or leakage. To retain the hot air, the system shall be insulated and heat shielded, as required.

3.35.4 Performance. The winterization kit shall be of such design that the following criteria can be met when the loader has been cold soaked for 24 hours at a temperature of minus 50° F:

(a) Engine start within 5 minutes after preheating engine coolant and battery box(s) for 90 minutes.
(b) Within 1 hour after personnel heater start, the heater shall have raised and stabilized the cab temperature to a desired plus 40° F in the area occupied by the operator's torso, seated.

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(c) Within 1 hour after personnel heater start, the defrosting system shall have cleared sufficient glassed area for safe operation of the loader and visibility of the loader bucket and work area normally seen when there is no frost on the windows.
(d) All loader controls shall be operational.
(e) The transmission and brake controls shall be functional.

3.35.5 Kit installation. The winterized loader shall be capable of performing loading operations.

3.36 Vehicle weight classification sign kit. The contractor shall apply the vehicle weight classification number to each vehicle utilizing a vehicle weight classification sign set (see 3.33). The set shall be bolted on the front of the loader in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the loader.

4. QUALITY ASSURANCE PROVISONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection. The first article (preproduction model) loader (see 6.2), furnished by the contractor shall be examined (see 4.3.1), and demonstrated (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and demonstration of the loader the following shall be performed:

(a) Service the loader with the lubricants specified.
(b) Break in the loader as prescribed by the contractor.

4.3.1 Preproduction examination. Prior to the preproduction demonstration, the loader shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection. Prior to the preproduction demonstration, the loader shall be examined and compared with this specification, with the contractor's published brochures, and with the complete Technical Information Package. Any discrepancy between the specification,

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brochures, or Technical Information Package and the preproduction model shall be cause for rejection. Prior to the preproduction demonstration; the calculations, the certifications, and certified test reports required by 4.5.2.19, 4.5.2.20 and 4.5.2.21 shall be examined. Incomplete, inaccurate, or otherwise unacceptable calculations, certifications, or certified test reports shall be cause for rejection.

TABLE II. Examination schedule.

Preproduction Model	Individual	Defect		Requirements Paragraph
x	x	101.	Safety items not as specified.	3.6
x	x	102.	Sound warning plate or decal (if required) not as specified.	3.6.1
x	x	103.	ROPS/FOPS mounting not as specified.	3.6.2
x	x	104.	Seat belt not as specified.	3.6.2
x	x	105.	Fenders not as specified.	3.6.3
x	x	106.	Horn not as specified.	3.6.4
x	x	107.	Backup alarm not as specified.	3.6.5
x	x	108.	Dimensions not as specified.	3.7.2, 3.20.1
x	x	109.	Cold start aid not as specified.	3.9
x	x	110.	Exhaust system not as specified.	3.11
x	x	111.	Fuel tank draining provisions not as specified.	3.11.3
x	x	112.	Engine side panels not as specified.	3.11.4
x	x	113.	Oil sampling valve not as specified.	3.11.5
x	x	114.	Transmission not as specified.	3.12.1
x	x	115.	Torque converter not as specified.	3.12.2
x	x	116.	Drive axles not as specified.	3.12.3
x	x	117.	Tires not as specified.	3.13
x	x	118.	Steering system not as specified.	3.14.1
x	x	119.	Braking system not as specified.	3.14.2
x	x	120.	Cab not as specified.	3.15
x	x	121.	Seat not as specified.	3.15.1
x	x	122.	Instruments and indicators not as specified.	3.15.2
x	x	123.	Controls not as specified.	3.15.3
x	x	124.	Boom kickout not as specified.	3.16

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TABLE II. Examination schedule.(con't)

Preproduction Model	Individual	Defect	Requirements Paragraph
x	x	125. Bucket positioner not as specified.	3.16
x	x	126. Electrical system not as specified.	3.17
x	x	127. Battery not as specified.	3.17.1
x	x	128. Alternator not as specified.	3.17.2
x	x	129. Lighting not as specified.	3.17.3
x	x	130. Blackout lighting not as specified.	3.17.4
x	x	131. Towing device not as specified.	3.18
x	x	132. Multipurpose bucket not as specified.	3.19.1
x	x	133. Counterweight and tire ballast not as specified.	3.19.2
x	x	134. Steering frame lock not as specified.	3.20
x	x	135. Shipping date plates not as specified.	3.23
x	x	136. Stenciling instructions not as specified.	3.24
x	x	137. Identification plates not as specified.	3.25
x	x	138. Instruction plates not as specified.	3.26
x	x	139. Treatment and painting not as specified.	3.27
x	x	140. Lubrication not as specified.	3.28
	x	141. Technical publications not as specified.	3.29
x	x	142. Special tools not as specified.	3.30
x	x	143. Toolbox not as specified.	3.31
x	x	144. Technical information package incomplete or not up to date.	3.32
x	x	145. Government-furnished equipment not installed.	3.33
x	x	146. Winterization kit not as specified.	3.35.1
x	x	147. Heater and heat exchanger not as specified.	3.35.2
x	x	148. Insulation not as specified.	3.35.3
x	x	149. Vehicle weight classification sign set not furnished.	3.36
x	x	150. Parts or components missing.	Section 3

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TABLE II. Examination schedule.(con't)

Preproduction Model	Individual	Defect	Requirements Paragraph
x	x	151. Assembly incomplete or incorrect.	Section 3
x	x	152. Cracked or broken parts or assemblies.	Section 3
x	x	153. Leaks in coolant, fuel or oil lines.	Section 3
x	x	154. Permanent deformation of any part or assembly.	Section 3
	x	155. Material not of same quality as used on preproduction model.	Section 3
x	x	156. Loader not as specified.	Section 3

4.3.2 Preproduction demonstration. Upon successful completion of the examination specified in 4.3.1, the preproduction model loader shall be subjected to demonstrations marked "X" in column 1 of table III. Failure of any demonstration shall be cause for rejection.

TABLE III. Demonstration schedule.

Preproduction Model	Individual	Demonstration	Demonstration Paragraph	Requirement Paragraph
X	X	Production run-in.	4.5.2.1	3.28
X	-	Sound level.	4.5.2.2	3.6.1
X	-	ROPS/FOPS removal.	4.5.2.3	3.6.2
X	-	Weight.	4.5.2.4	3.7.2, 3.20.1
X	-	Tipping load.	4.5.2.5	3.10.2
X	-	Lift capacity.	4.5.2.6	3.10.3
X	-	Breakout force.	4.5.2.7	3.10.4
X	-	Clam closure force.	4.5.2.8	3.10.5
X	-	Drawbar pull.	4.5.2.9	3.10.6
X	-	Travel speed.	4.5.2.10	3.10.7
X	-	Cycle time.	4.5.2.11	3.10.8
X	-	Clearance circle.	4.5.2.12	3.10.9
X	-	Stability.	4.5.2.13	3.10.10
X	-	Cold test	4.5.2.19	3.35.4
X	-	Installation test	4.5.2.20	3.35.5
X	-	Fording.	4.5.2.14	3.10.11
X	-	Braking system.	4.5.2.15	3.14.2
X	-	Slinging provisions.	4.5.2.16	3.21
X	-	Tiedown provisions.	4.5.2.17	3.22
X	-	Construction tasks.	4.5.2.18	3.8, 3.11.3

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4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the demonstration specified in 4.4.2, each loader shall be examined for the defects marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual demonstration. Each loader shall be demonstrated as specified in column 2 of table III. Failure of any demonstration shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table II.

4.5.2 Demonstration. Demonstrations shall be conducted in accordance with table III.

4.5.2.1 Production run-in. Start and operate each loader for not less than 1/2 hour to assure operation of all components and to allow for minor adjustments. Start and stop the engine not less than five times. Operate all bucket controls for five minutes. Hold each hydraulic control in the actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the loader at least two times through a figure eight steering course at the maximum safe speed. Make not less than five abrupt panic stops from a speed not less than 20 mph using the service brakes. Release and reapply the parking brakes not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hyraulic fluid shall be cause for rejection of the loader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that loaders previously accepted are similary defective unless evidence, satisfactory to the contracting officer, is furnished by the contactor that they are not similarly defective.

4.5.2.2 Sound level. The contractor shall perform the sound level tests in accordance with SAE J88 and SAE J919. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.6.1 shall constitute failure of this demonstration.

4.5.2.3 ROPS/FOPS removal and replacement. The removal and replacement of the ROPS/FOPS structure shall be demonstrated following procedures in the contractor's manual. Permanent deformation or damage to any part of the ROPS/FOPS structure or its mounts, or inability to meet the requirements of 3.6.2 and 3.20 shall constitute failure of this demonstration.

4.5.2.4 Weight. The weight of the loader shall be measured with a scale calibrated within six (6) months of the demonstrations. Inability to meet the requirements of 3.7.1 and 3.20.1 shall constitute failure of this demonstration.

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4.5.2.5 Tipping load. The tipping load of the loader shall be measured to prove conformance to 3.10.1 and 3.10.2. Nonconformance to 3.10.1 and 3.10.2 shall constitute failure of this test.

4.5.2.6 Lift capacity. The lift capacity at the carry position and the lift capacity to maximum height shall be measured to prove conformance to 3.10.3. Nonconformance to 3.10.3 shall constitute failure of this test.

4.5.2.7 Breakout force. The breakout force of the loader shall be measured to prove conformance to 3.10.4. Nonconformance to 3.10.4 shall constitute failure of this test.

4.5.2.8 Clam closure force. The clam closure force of the loader shall be measured to prove conformance to 3.10.5. Nonconformance to 3.10.5 shall constitute failure of this test.

4.5.2.9 Drawbar pull. The drawbar pull of the loader shall be measured to prove conformance to 3.10.6. Nonconformance to 3.10.6 shall constitute failure of this test.

4.5.2.10 Travel speed. The travel speed of the loader shall be measured to prove conformance to 3.10.7. Nonconformance to 3.10.7 shall constitute failure of this test.

4.5.2.11 Cycle time. The raising time, lowering time, and dump time of the loader shall be measured to prove conformance to 3.10.8. Nonconformance to 3.10.8 shall constitute failure of this test.

4.5.2.12 Clearance circle. The clearance circle of the loader shall be measured to prove conformance to 3.10.9. Nonconformance to 3.10.9 shall constitute failure of this test.

4.5.2.13 Stability. The stability of the loader shall be measured to prove conformance to 3.10.10. Nonconformance to 3.10.10 shall constitute failure of this test.

4.5.2.14 Fording. The fording ability of the loader shall be demonstrated to prove conformance of 3.10.11. Nonconformance to 3.10.11 shall constitute failure of this test.

4.5.2.15 Braking systems. The braking systems of the loader shall be tested to prove conformance to 3.14.2. Nonconformance to 3.14.2 shall constitute failure of this test.

4.5.2.16 Slinging provisions. The slinging provisions shall be demonstrated to prove conformance to 3.21. Nonconformance to 3.21 weld failure, or permanent deformation of the loader or the slinging provisions shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

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(a) Lift the loader and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the loader. Spreader bars are permitted. Determine the force and angles of application for each provisions. Measure the clearance between each sling and the loader.
(b) Restrain the loader by anchoring the main frame and subject each slinging provisions to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for a minimum of 90 seconds.

4.5.2.17 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.22. Apply a force equal to the limit load as specified in MIL-STD-209 to each of the tiedown provisions. Weld failure or permanent deformation of the loader or the tiedown provisions shall constitute failure of this demonstration.

4.5.2.18 Construction tasks. The loader shall be demonstrated as follows for a total of 100 hours to prove conformance to 3.8 and 3.11.3:

(a) Using bank run gravel or building sand demonstrate for 35 hours loading (from stockpiles) dump trucks having a sideboard height of not less than 80 inches above ground level. Approximately one-half of this operation shall be conducted with the load bottom dumped.
(b) Using bank run gravel or building sand demonstrate for 35 hours the loader's ability to move stockpiles with capacity loads a distance of not less than 100 feet. Approximately one-half of this operation shall be conducted with load dumped.
(c) Demonstrate for 15 hours the loader's ability to excavate a pit in undisturbed soil using the bucket closed.
(d) Demonstrate for 15 hours the loader's ability to excavate a pit in undisturbed soil using the open bucket as a dozer.

Nonconformance to 3.8 and 3.11.3 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this demonstration.

4.5.2.19 Cold tests. The loader shall be cold soaked in a temperature controlled chamber to a temperature of minus 50° F and demonstrated as follows to prove conformance to 3.35.4.

(a) Preparation.

1. Insure that the loader has been serviced with arctic coolant, fuels, lubricants and hydraulic oils.
2. Service Swingfire heater with arctic gasoline.
3. Place loader in temperature controlled chamber.
4. Instrument the loader to record as a minimum the following temperatures:
   a. Exhaust, engine heater outlet.
   b. Air, battery box(s), inlet, outlet, and all aides.
   c. Air, operator's compartment, operator torso (seated positioned).

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d. Air, operator's compartment, floor, rook, sides, (midway to top).
e. Air, hot water heater outlet.
f. Electrolyte, all battery(s).
g. Coolant, engine block inlet.
h. Coolant, engine block outlet.
i. Oil, engine sump and gallery.

(b) Procedure.

1. Subject loader to a stabilized temperature of minus 500° F for 24 hours.
2. Bring Swingfire heater into cold chamber. Connect heater to loader electrical power source and start heater. (After heater start disconnect from power source).
3. Insert heater into heat exchanger and preheat engine coolant and battery(s) for 90 minutes.
4. Record temperatures continuously or at intervals not greater than five minutes.
5. Remove heater from heat exchanger and shut off motor.
6. Start engine from loader battery(s).
7. Start personnel heater and defrost fans, 30 minutes after engine start.
8. One hour after heater start, record operator compartment air temperatures and check defrost.
9. Operate loader steering and bucket controls for 10 minutes, within the dimensional limits possible within the test chamber.
10. Engage and operate transmission and brake system as dimensional limits of test chamber permits.

Nonconformance to 3.35.4 shall constitute failure of this demonstration.

4.5.2.20 Installation test. The purpose of this test is to insure the proper fit, form and function of the winterization kit and that no operational or safety deficiencies exist after installation of the kit on the loader for which it is intended. The loader shall be demonstrated for 10 hours to prove conformance to 3.35.5.

4.5.2.21 Calculation. The contractor shall furnish calculations to prove conformance to the bucket capacity requirements of 3.19.1 and the air transportability requirements of 3.20.1. Dimensions used in the calculations shall be verified by measurement. Nonconformance to 3.19.1 or 3.20.1 shall be cause for rejection.

4.5.2.22 Certification. The contractor shall certify in writing to the contracting officer that the loader complies with the starting and operating temperature requirements of 3.9, the engine requirements of 3.11, the horsepower requirement of 3.11.1, the steering requirement of 3.14.1, and the rail shipment requirements of 3.20. If certification is not supplied, these requirements shall be demonstrated. Nonconformance shall constitute failure of this demonstration.

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4.5.2.23 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS/FOPS requirements of 3.6.2, and electromagnetic radiation of 3.17.5. If certified test reports are not available, these requirements shall be demonstrated. Nonconformance shall constitute failure of these demonstrations.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with MIL-T-3351 (for level A or B) and MIL-STD-1188 (for commercial).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each complete loader, components and attachments shall be preserved in accordance with the level A preservation and packaging requirements of MIL-T-3351.

5.1.2 Level B. Each complete loader, components and attachments shall be preserved in accordance with level B preservation and packaging requirements of MIL-T-3351.

5.1.3 Commercial. Each complete loader and all items issue with the loader shall be suitability preserved and packed in accordance with MIL-STD-1188 for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservation coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent anti-freeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the loader in operation.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

5.2.2 Level B. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

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5.2.3 Commercial. Each complete loader, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188. Commercial packing must protect items against physical and environmental damage during shipment, handling and storage.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The loader is intended to be used primarily for excavating, digging, loading, stripping, leveling, stockpiling, and transferring of earth and aggregrate.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame for submission of preproduction model (see 3.3).
(c) Time frame for completing and submitting the vehicle classification data sheet (see 3.23.1)
(d) Technical publications required (see 3.29).
(e) Time frame for submission of Technical Information Package (see 3.32).
(f) When Basic Issue Items are to be furnished (see 3.34).
(g) When winterized loaders are required and number of loaders to be furnished (see 3.35).
(h) Time frame required for submission of winterized loader for examination and demonstration (see 3.35).
(i) When Swingfire heaters are required and number of heaters to be furnished (see 3.35.2).
(j) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.33.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

Custodian:
Army - ME

User Activities:
Army - AT, CE

Preparing activity:
Army - ME

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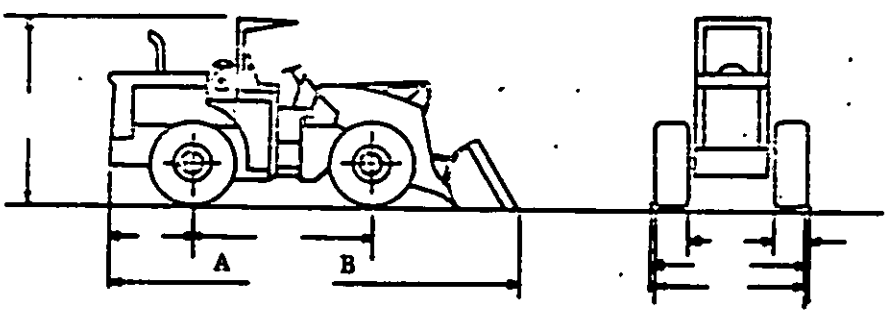
VEHICLE CLASSIFICATION DATA
LOADER
DATA REQUIRED FOR IDENTIFICATION AND LOAD CLASSIFICATION OF SUBJECT VEHICLE

Load Class		
E	C	H

NOMENCLATURE:

Mfr.	Model No.
NSN	Agency / Stock No.

Traveling Load Category		Total Weight W	Pay Load
Empty	E.		X
Loaded	C		



Traveling Load Category	LOADS ON		Front Tires			Rear Tires		
	A	B	No/ axle	size	psi	No/ axle	size	psi
Empty								
Loaded								

All loads in short tons.
All loads to include weight of crew and appurtenances.
Class No's. for vehicles over 3,000 lbs.
Longitudinal dimensions in feet.
Vertical and transverse dimensions in inches.

FIGURE 1. Vehicle classification data sheet.

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-L-52385C(ME) Loader, Scoop Type, DED, 4X4, Articulated Frame Steer, 2-½ Cu. Yd., Commercial Constr Equip (CCE)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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